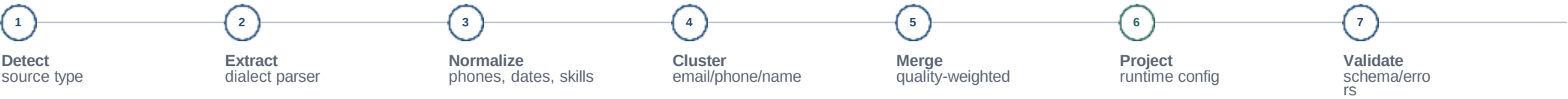


Multi-Source Candidate Data Transformer

Technical design one-pager | Dasoju Shashank | shashankdasoju111@gmail.com | Eightfold Engineering Intern Assignment

Design intent: produce one trustworthy canonical profile per candidate; never invent values; keep every accepted value traceable.

Pipeline breakdown



CANONICAL SCHEMA AND FORMATS

- **Identity:** candidate_id is UUIDv5 from stable keys; full_name preserves the best display value.
- **Contact:** emails are lowercase/RFC-style checked; phones normalize to E.164 and dedupe by normalized key.
- **Location:** location = {city, region, country}; country → ISO-3166.
- **Links:** links split into linkedin, github, portfolio, and other.
- **Career data:** headline, years_experience, experience[{company,title,start,end,summary}], education[{institution,degree,field,end_year}].
- **Trust fields:** skills[{name, confidence, sources[]}], provenance[{field, source, method}], overall_confidence in [0,1].

NORMALIZATION POLICY

- Dates keep the most precise known form; month-granular values use YYYY-MM and unknown dates remain null.
- Skills use a small alias map before dedupe, so 'js', 'JavaScript', and 'javascript' collapse deterministically.
- No extractor fabricates missing values; malformed sources produce empty records plus warnings.

SOURCE COVERAGE

- **Structured:** recruiter CSV and ATS JSON are parsed through dialect-specific adapters, then mapped into the canonical model.
- **Unstructured:** resume/TXT extraction is conservative and regex-led; GitHub URL support enriches links and public profile fields when available.
- **Malformed input:** parser-level failures are isolated to that source and surfaced as warnings, not process crashes.

MERGE AND CONFLICT RESOLUTION

- **Identity clustering:** email has highest authority, then phone; fallback is normalized name + country.
- **Scalar fields:** choose the value with the highest dynamic source quality score, based on completeness and parse confidence.
- **List fields:** union, normalize, and dedupe emails, phones, links, and skills.
- **Experience:** group by normalized company; keep the most precise date range and concatenate unique summaries.
- **Conflicts:** deterministic tie-break by source order, with provenance marking the selected source and method.

CONFIDENCE MODEL

- Source quality is dynamic rather than hard-coded by file type: completeness, parse success, and field specificity matter.
- Skill confidence = frequency * 0.5 + average source quality * 0.5.
- Overall confidence combines coverage and average accepted-field quality, bounded to [0,1].

VALIDATION AND FAILURE MODE

- Projection validation runs after remapping, so custom outputs are checked against the requested shape, not only the internal schema.
- Required-field failures become explicit ProjectionError entries when on_missing = error.
- Default behavior favors honest nulls and an error list over silent bad values.

RUNTIME OUTPUT CONFIG

- The engine first builds a CanonicalProfile; a Projection Gateway reshapes it without code changes.
- fields[].path defines the output key; fields[].from reads canonical paths such as emails[0], skills[].name, location.country.
- Per-field normalize supports E164, lowercase, canonical, or pass-through.
- include_confidence and include_provenance are toggles; ignored_sources can veto known-bad input types.
- on_missing supports null, omit, or error; post-projection validation checks required fields and types.

EDGE CASES AND SCOPE

- **Missing/empty source:** skip safely and return warnings.
- **Multiple candidates:** CSV rows become separate clusters unless contact keys connect them.
- **Name capitalization conflicts:** lowercase for matching; best-quality original form for display.
- **Unknown ATS fields:** ignored instead of guessed.
- **Descope:** LinkedIn scraping due ToS, OCR for scanned PDFs, async ingestion, and ML NER.

SCALE AND TRADEOFFS

- Identity clustering is graph-based over normalized keys, avoiding pairwise fuzzy matching for the common path.
- The CLI is intentionally thin; correctness, provenance, and graceful degradation get priority over UI polish.
- More advanced ranking, OCR, and async ingestion are natural follow-ups after the deterministic core is stable.

Implementation: Python 3.12, pydantic, phonenumbers, pycountry, pypdf, python-docx, requests, reportlab. **CLI:** python cli.py --sources [--config] [--output]. **Demo:** ./run_demo.sh. **Tests:** pytest tests/ -v.